## 1.0 Background of the Study

Personal safety remains a significant concern for individuals who travel alone, commute through unfamiliar environments, or move through areas where immediate assistance may not be readily available. Conventional safety applications usually involve the user manually sharing their location, pressing an emergency button, or making a call they perceive a threat. However, these approaches may be ineffective when the user is unable or unwilling to interact with the application during an emergency.

Location-based technologies provide an opportunity to improve personal safety by enabling continuous monitoring of a user's movement. By comparing a user's current location with an intended route, it is possible to identify unexpected changes in movement that may indicate an unusual or potentially unsafe situation.

The proposed system consists of a web platform and a mobile application for emergency monitoring that enables users to register trusted emergency contacts, specify a destination, select mode of transportation and initiate a journey. During the journey, the system monitors the user's location and compares the user's movement with the planned route. The designated emergency contacts are provided with detailed information about the user's route and destination.

To initiate a journey, the user specifies a destination and selects the intended method of transportation. The system then establishes a planned route based on the selected journey parameters. During the journey, the user's movement is monitored against the planned route, while designated emergency contacts are provided with the user's route and destination information.

The system also accounts for environmental constraints that may affect route deviation detection. Where roadblocks or other relevant route obstructions are not automatically detected or are inaccurately represented by available route information, the user may manually report the obstruction through the application. Such manually reported roadblocks are explicitly identified as user-provided information which is also shown to the emergency contacts.

When a significant deviation from the planned route is detected, the system sends an initial notification to the emergency contacts. The system then applies a user-configurable deviation threshold during which the deviation may be resolved. If the deviation persists beyond this threshold, the system enters an escalation state and presents the user with an emergency alert containing a user-configurable cancellation window. If the alert is not cancelled within the specified period, the system sends the emergency notification to the designated contacts and initiates audio recording from the user's device. The recorded information can then be transmitted to the emergency contacts, subject to device permissions and network availability.

The system therefore combines route tracking, deviation detection, automated notification, time-based escalation, and emergency evidence capture into a single safety solution. Its purpose is to provide an additional layer of protection by reducing dependence on manual emergency activation.

## 1.1 Statement of the Problem

Existing personal safety applications often depend on the user manually initiating an emergency alert. This presents a limitation in situations where the user is unable to access or operate the application.

In addition, conventional location-sharing systems primarily communicate a user's current location without determining whether the user is following the intended route. Consequently, an unexpected change in movement may not be identified or communicated to trusted contacts in a timely manner.

Another challenge is that not every route deviation constitutes an emergency. Traffic conditions, road closures, navigation errors, or personal decisions may result in temporary deviations. A safety system that immediately classifies every deviation as an emergency may therefore generate unnecessary alerts.

Furthermore, basic emergency notifications may provide limited information about the circumstances surrounding an incident. Location information alone may not be sufficient for emergency contacts to understand what is happening or respond appropriately. There is therefore a need for an escalation mechanism that first alerts designated contacts, provides a configurable period for the deviation to be resolved, and subsequently gives the user an opportunity to cancel an emergency escalation before an emergency notification and automatic recording are initiated.

## 1.2 Significance of the Study

The proposed system is significant because it introduces automated route monitoring into personal safety management. Rather than relying solely on manual emergency activation, the system can identify significant deviations from an intended route and initiate the alert process automatically.

The system provides emergency contacts with greater awareness of a user's journey by communicating relevant route and destination information and notifying them when an unexpected deviation occurs. The configurable escalation period also allows temporary deviations to be distinguished from persistent deviations before an emergency alert is triggered.

The system also provides a mechanism for incorporating user-observed environmental conditions into journey monitoring. Users can manually report roadblocks that are not adequately represented by available route information, allowing such information to be displayed to designated emergency contacts and considered within the context of the user's journey. The separation between initial deviation notification, escalation, cancellation, and emergency notification is also significant because it provides multiple stages for assessing an abnormal journey event rather than treating every deviation as an immediate emergency.

The automatic recording functionality provides an additional source of contextual information during an escalated event. Where supported, an audio recording may assist emergency contacts in understanding the circumstances surrounding the alert.

## 1.3 Justification of the Study

The study is justified by the limitations of safety systems that depend entirely on user-initiated emergency actions. During a potentially dangerous situation, a user may not have sufficient opportunity to interact with a mobile device. Automated route monitoring provides a mechanism through which a possible emergency can be identified without requiring immediate manual intervention.

The proposed system also addresses the possibility of false or unnecessary emergency escalation by separating route-deviation detection from emergency declaration. Once a significant deviation is detected, an initial notification is sent to the designated contacts and a configurable period is provided for the deviation to be resolved. If the deviation persists, the user is presented with an emergency alert and a configurable cancellation window. Emergency notification to the designated contacts and automatic audio recording occur only when the cancellation period expires without user intervention. This layered approach provides additional opportunity to distinguish genuine emergency situations from temporary or intentional route changes.

The inclusion of automatic audio recording further strengthens the system by providing contextual information in addition to location data. This can make an escalated alert more informative to the designated emergency contacts.

The project is also justified as a practical application of modern web, mobile and location-based technologies to a real-world safety problem. It provides an opportunity to design, implement, integrate, and evaluate a system involving multiple software and device-level components.

## 1.4 Aim and Objectives of the Study

The primary aim of this research project:
*is to design and implement a web-based and mobile-based emergency route monitoring system that monitors a user's journey based on a selected transportation method, detects significant route deviations, notifies designated emergency contacts, and captures additional contextual information during an escalated event.*

The specific objectives are to:

1. Develop a web and mobile system for user registration and management of designated emergency contacts.
2. Implement journey configuration through destination selection and transportation-method selection.
3. Implement route generation and continuous location monitoring during an active journey.
4. Develop a mechanism for detecting significant deviations from the planned route.
5. Implement automatic notification of designated emergency contacts when a significant route deviation is detected.
6. Implement a user-configurable time threshold for determining deviation escalation.
7. Implement an emergency escalation mechanism that provides the user with a configurable cancellation window before emergency notification is initiated.
8. Automatically notify designated emergency contacts and initiate audio recording when an escalated emergency is not cancelled within the configured cancellation period.
9. Implement a mechanism for users to manually report roadblocks or route obstructions that are not adequately represented by available route information and make such reports visible to designated emergency contacts.
10. Test and evaluate the system under normal journey, route deviation, roadblock, escalation, cancellation, notification, and audio-recording scenarios.

## 1.5 Methodology Overview of the Study

The system adopts an iterative software development methodology consisting of requirements analysis, system design, implementation, testing, and evaluation.

The requirements analysis phase identifies the functional and non-functional requirements of the proposed system, including user management, emergency contact management, route creation, location monitoring, deviation detection, notifications, escalation, and audio recording.

The system design phase defines the architecture and behaviour of the application. This includes the design of the database, user interface, route-monitoring process, notification workflow, and emergency escalation mechanism. Appropriate system modelling techniques are used to represent the proposed solution.

The implementation phase involves developing and integrating the identified system components. The application obtains the user's location, monitors movement during an active journey, detects deviations from the planned route, and executes the appropriate notification and escalation procedures.

The testing phase evaluates the functionality and reliability of the system using defined test cases and simulated usage scenarios. These include normal route completion, temporary route deviation, persistent route deviation, notification delivery, and automatic audio capture. Testing will also verify the complete emergency state sequence, including normal route progression, significant route deviation, initial

contact notification, deviation timeout, emergency escalation, cancellation within the configured window, expiry of the cancellation window, emergency notification, and automatic audio recording. Roadblock reporting and the visibility of user-reported roadblocks to designated emergency contacts will also be tested.

The evaluation phase assesses the behaviour of the completed system against the defined objectives and requirements.

## 1.6 Scope of the Study

The system will allow users to register designated emergency contacts, specify a destination, select a method of transportation, and initiate a journey. Based on these journey parameters, a planned route will be established and monitored throughout the active journey. Designated emergency contacts will have access to the user's relevant journey information, including the planned route and destination, and will receive notifications when defined route-deviation events occur.

The system will detect significant deviations from the planned route and issue an initial notification to the designated emergency contacts. A user-configurable deviation threshold will determine how long the deviation may persist before escalation. If the deviation continues beyond the threshold, the system will present an emergency alert to the user with a user-configurable cancellation window. If the emergency is not cancelled within the specified period, the system will send the emergency notification to the designated contacts and initiate audio recording from the user's device.

The system will also provide a mechanism for users to manually report roadblocks or other relevant route obstructions when such environmental constraints are not automatically detected or are inadequately represented by available route information. User-reported roadblocks will be identified as manually entered information and will be made visible to emergency contacts.

The recorded audio and relevant emergency information may be transmitted to the designated contacts, subject to device permissions, network availability, and other technical limitations.

The study does not include direct physical intervention, guaranteed emergency response, medical diagnosis, crime prediction, or autonomous deployment of emergency personnel. The effectiveness of location monitoring, notification delivery, and audio transmission may also be affected by factors such as GPS accuracy, network availability, device permissions, battery level, and device capabilities.
Video recording is outside the primary scope of the study unless it is successfully implemented within the available development environment and project constraints.

## 1.7 Organisation of the Project

This project is presented in five chapters.

Chapter One - Introduction presents the background of the study, statement of the problem, significance and justification of the study, aim and objectives, methodology overview, scope, and organisation of the project.

Chapter Two - Literature Review presents introduction, concept review, theoretical review, review of existing systems, research gap, and summary of the literature review.

Chapter Three - Software Requirement Engineering presents introduction, overview of the proposed system, requirement analysis, software requirement specification, requirement modelling, and chapter summary.

Chapter Four - focuses on the implementation of the system, detailing the development tools, programming environment, code structure and screenshots of the working system.

Chapter Five - concludes the project, summarising the work done, stating the challenges encountered and suggesting possible future improvements to enhance the system's functionality and performance.